

The influence of political disagreements and corruption on state health leader turnover during the COVID-19 pandemic in Brazil

Political
disagreements
and corruption

Received 17 January 2024
Revised 12 April 2024
Accepted 7 May 2024

Lorena G. Barberia

Ciência Política, Universidade de São Paulo, São Paulo, Brazil

Gilmar Masiero

Administração, Universidade de São Paulo, São Paulo, Brazil

Iana Alves de Lima

EAESP, Getúlio Vargas Foundation, São Paulo, Brazil

Luciana Santana

Ciência Política, Universidade Federal de Alagoas, Maceio, Brazil, and

Tatiane C. Moraes de Sousa

Ciência Política, Universidade de São Paulo, São Paulo, Brazil

Abstract

Purpose – Governments faced formidable challenges in coordinating public health responses to the COVID-19 pandemic. This study aims to enhance the understanding of effective organizational leadership during crises by investigating the factors influencing the turnover of health leaders during the COVID-19 pandemic in Brazil.

Design/methodology/approach – Using primary data encompassing all appointments and dismissals of federal and state health secretaries, this paper conducted a quantitative analysis of the relational and reputational factors that contributed to leader turnover during the COVID-19 pandemic. This paper also examined whether leaders' management and public health experience increase the duration of tenure.

Erratum: It has come to the attention of the publisher that the article, Barberia, L.G., Masiero, G., Alves de Lima, I., Santana, L. and Moraes de Sousa, T.C. (2024), "The influence of political disagreements and corruption on state health leader turnover during the COVID-19 pandemic in Brazil", *Leadership in Health Services*, Vol. ahead-of-print No. ahead-of-print. <https://doi.org/10.1108/LHS-01-2024-0008>, did not include the corrected Figure 4a and Table 3 in the article. This correction was missed by the production team during the typesetting process. This error has been corrected in the online version of the article. The publisher sincerely apologises to readers for any inconvenience caused.

The researchers thank Natalia de Paula Moreira, Marcela Zamudio, Vitória Lopes and Aryanne Alcântara for invaluable research assistance. This work was supported by the Sao Paulo Research Foundation (FAPESP) [grant numbers 2021/08772-9 and 2022/10997-1]. The funding received by the researchers did not directly contribute to the study design, data collection, analysis, data interpretation, manuscript writing and the decision to submit the manuscript for publication. The authors declare no competing financial interests.



Findings – States encountered significant challenges in retaining experienced and effective leadership during the health emergency, primarily due to political conflicts in policymaking and, to a lesser extent, allegations of corruption. Furthermore, leaders with expertise in public health were found to be less likely to be removed from office. However, managerial experience did not prolong the tenure of state health secretaries during the emergency.

Research limitations/implications – Since most health leaders have public health and management experience, the contributions of each factor to the duration of a secretary's tenure are difficult to separate and analyze separately.

Practical implications – This study provides empirical insights into what factors drive health leader turnover during major health emergencies.

Social implications – During major health emergencies, health leaders often strongly disagree with elected officials on the response. This paper tests how crisis leadership theories help explain state health leaders' duration in one of the world's largest public health systems during the COVID-19 pandemic. This paper finds that policy disagreements contributed to significant turnover.

Originality/value – To the best of the authors' knowledge, this paper is the first that is aware of that uses novel primary data on public health executive leader characteristics and turnover causes in the context of the COVID-19 pandemic. It provides empirical evidence contributing to the crisis leadership literature by examining health leader turnover in one of the world's largest public health systems.

Keywords Public health, Politics, Health leadership competencies, Crisis leadership

Paper type Research paper

Introduction

The COVID-19 pandemic represents one of the most significant global tests of health leadership (Geerts *et al.*, 2021). In countries where the emergency coincided with a highly politicized and polarized setting, there are indications of high turnover in key government posts (Ramírez de la Cruz *et al.*, 2020; Park and Fowler, 2021). In Brazil, for example, Brazilian President Jair Bolsonaro strongly contested that a robust public health response was necessary, arguing instead that economic growth and prosperity concerns should prevail in reaction to the COVID-19 emergency. During his four-year term, the Brazilian health minister changed hands four times, and three of these turnovers were in the first year of the pandemic when the situation was in a heightened emergency stage.

The difficulties of forging and maintaining strategic alignments during pandemic crises and, most notably, in a polarized and politicized environment were not only a significant challenge for senior government posts at the national level (Pereira *et al.*, 2020). Like the volatility at the ministry level, there was also significant turnover in leadership in states during the COVID-19 pandemic. There is limited understanding of the factors that drive alternations in senior public managers (Cregård, Corin and Skagert, 2017). In the case of health emergencies, these lacunae are even more significant.

There are important reasons to study public leadership turnover at the state level in federations and even more so in the context of the COVID-19 pandemic. While the Brazilian federal government encountered numerous difficulties in retaining a health minister and battling to loosen non-pharmaceutical interventions in the judiciary, governors and their health secretaries assumed a central role in coordinating the response to the tragedy across the federation (Pereira *et al.*, 2020). In the early phases of the emergency, state governments increased intensive care units in the public health system or emergency field hospitals, developed rapid response systems to transfer cases from metropolitan to other locations where there was a lower demand for health services and vice versa, and even resorted to routing patients with COVID-19 to the private system where there was additional intensive care unit availability (Cruz, 2020).

The leaders of states, such as São Paulo, were also critical in building emergency field hospitals, supporting clinical trials, securing agreements to procure and produce vaccines in

2020 and organizing immunization in 2021 (Fonseca *et al.*, 2021). This mobilization of human and physical resources by state governments was especially noteworthy considering the significant challenges faced by most administrations considering that infrastructure in the Unified Health System (Sistema Único de Saúde - SUS) was unevenly distributed (Pereira *et al.*, 2020) and there had been systemic cuts in health spending in the years prior to the onset of the COVID-19 pandemic (Massuda *et al.*, 2018). Thus, this study contributes to research on organizational leadership that effectively manages health emergencies.

In this study, we adopt the framework proposed by Andrus *et al.* (2019) on the causes of top executive turnover. We show that relational and, to a lesser degree, reputational shocks increased the likelihood of executive exit. We also consider the influence of managerial and public health experience in moderating how these shocks affect turnover. The remainder of the paper is organized as follows. The next section describes how our study contributes to the literature on leadership turnover in healthcare. We then describe the data collection process and explain the methods we employed to investigate which factors contributed to health leadership turnover at the state level. The main results are presented, followed by a conclusion and discussion of findings, emphasizing practical implications.

Theory

A significant strand of the literature documents associations between turnovers in leadership and how these changes impact the strategic alignment of organizations that deliver healthcare (Anderson and McDaniel, 2000; Alexander *et al.*, 2003; Ginter *et al.*, 2018), the distinct challenges for public health (Fraser *et al.*, 2017; Allen *et al.*, 2018) and for national public health systems (Siddiqi *et al.*, 2009). This literature specifically emphasizes the role of leadership in ensuring strategic alignment during crises. Boin *et al.* (2005) and health emergencies (Geerts *et al.*, 2021). Research has focused on identifying how strategies and organizational structures changed during these events, considering the substantial and immediate threat (e.g. SARS-CoV-2 – severe acute respiratory syndrome coronavirus 2).

Our study is directly related to the literature on how leadership turnover has impacted the effective response of public health to the COVID-19 pandemic, given the disease's burden on the healthcare system. Leaders had to quickly undertake a massive effort to reallocate healthcare professionals, provide diagnostic testing and medications, procure vaccines, immunize the population and deliver the needed increase in capacity to handle a new disease while maintaining existing programs. Research has examined the extent to which a health organization's purpose endured (Cortez *et al.*, 2021; Haldane *et al.*, 2021; Ezzati *et al.*, 2023). For example, research efforts have assessed if public health ministries remained focused on their long-term goals of preventing disease, promoting health and prolonging life among the population (Kooli, 2021). In these studies, turnover in leadership is recognized as an opportunity that comes at a cost, as the transition period can impede an organization from responding effectively.

Haque (2021) hypothesizes that COVID-19 is negatively associated with organizational stability and encourages research to test this hypothesis across different contexts and types of organizations. Furthermore, in the case of developing countries, Ramírez de la Cruz *et al.* (2020) argue that limited professionalization, particularly in less developed cities, corruption, scarce resources, national government interventions and political risk aversion contribute to raising transaction costs for public managers, suggesting that turnover rates will increase during the pandemic.

In the case of Brazil, the continuous replacements of the minister of health and the senior leadership team resulted in the absence of national coordination and guidance (Barberia *et al.*, 2021a). As this leadership vacuum was combined with the restricted understanding of the disease's life cycle and significant resource limitations, subnational public health leaders

pursued different management strategies to meet the demand for COVID-19 patient care (Gomes *et al.*, 2021).

Several studies have described how this adverse context contributed to the challenges faced by hospitals (Beck da Silva Etges *et al.*, 2021) and municipalities (Bousquat *et al.*, 2022). These studies have yet to identify whether specific management traits and alliances decrease the likelihood of a significant turnover in health leadership during an emergency or whether the volatility in leadership was higher in periods in which there were leadership changes at the national level in the Ministry of Health.

Our research follows the literature that argues that relational and reputational events drive voluntary and involuntary individual-level turnover. We follow Andrus *et al.* (2019), who argue that relational and reputational shocks increase the likelihood of executive exit. This literature has been advanced in management, political science and public administration, and we adapt both concepts to analyze turnover in senior leadership in public health management.

Relational shocks refer to disagreements between state health secretaries and the governor or the Ministry of Health. Public health leaders work at a strategic level negotiating with political-administrative managers to make decisions. A state health secretary has a complex assignment. During respiratory health emergencies, the multiple pressures on leadership are further compounded (Boin and 't Hart, 2010). Leaders had to pursue competing objectives, choosing whether to prioritize purchasing costly lifesaving ventilators or less urgent medications while responding to demands sometimes driven by politics and ideological differences and, to a lesser extent, the threat itself. Often, secretaries must defend positions publicly that they privately would not endorse. When there is a public or internal conflict in policy, there is an increased likelihood that a governor will remove the secretary or that he/she will resign.

Politics and election cycles drive another type of relational shock. Health leaders are also politicians who are keenly interested in their political survival. As prominent actors, parties and local political networks will often nominate secretaries of health to run for office in municipal, state and national elections. As the campaign period begins, a secretary is more likely to leave office, and following an electoral defeat, these same leaders will often return to their posts.

A reputational shock, in contrast, can be considered an event that leads to the questioning of the leadership and management capacities of the public health leader. Studies show that reputational effects, such as the public reporting of performance based on benchmarking, improve public health service delivery at the national and subnational levels (Bevan *et al.*, 2019). We consider accusations concerning ethics violations, corruption and fraud allegations as types of reputational shocks that pressure leaders to resign or be removed (Zahra *et al.*, 2007).

We posit that relational (e.g. conflicts between the secretary of health and the governor and/or federal government) and reputational shocks (e.g. allegations of fraud, corruption and ethics violations) increase turnover. Two hypotheses follow:

H1. Relational shocks increase an executive's likelihood of exiting during a health emergency.

H2. Reputational shocks increase the likelihood of exit during a health emergency.

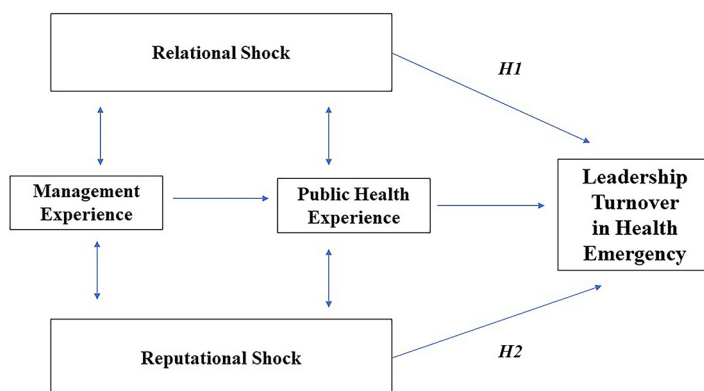
There are important reasons to consider that a state health secretary's medical background and managerial experience will contribute to the duration of secretaries in their posts during a health emergency. Research shows technocrats are at a lower risk of exit than partisan ministers (Camerlo and Perez-Liñan, 2015). In the face of a significant conflict, loyalty is the costliest, and voicing comes with risks whose consequences are uncertain. In these

situations, exit becomes an attractive means by which ambitious leaders can keep the “revolving door” in the public sphere in operation. Our study considers technical expertise, including academic training in medicine, health and related areas, or extensive job experience in the health sector as factors that affect turnover. When secretaries have more qualifications in the medical area and managing organizations, they will be less likely to exit. However, allegations of corruption and fraud during a major public health emergency will spark public outrage and pressure leaders to resign. Figure 1 summarizes our theoretical framework, showing how relational and reputational factors affect turnover and the contribution of public health and managerial experience to these trends.

Data and methods

We collected primary data on all appointments and dismissals of federal and state health secretaries between January 1, 2019, and December 31, 2022. The start date is when the president and state governors elected in 2018 in Brazil assumed office, and the end date is when their terms concluded. Within this period, we focused on the onset of the health emergency (February 1, 2020) to the end of the term of office of secretaries appointed by the governors elected in 2018 (December 31, 2022). To compare the duration of these health secretaries, we also analyzed the turnover of leaders appointed under previous health emergencies from 1995 to 2018.

For all appointees, we collected data regarding their time in the office (number of days), their reason for dismissal (if any), gender, professional work background in health care, public management or public health activities, including public health and management academic training and previous employment. We also collected data on the political party affiliation of the secretary, governor and the annual governor’s alignment with president Bolsonaro between 2019 and 2022. The data was collected through a triangulation of information using online sources: state health secretaries’ websites, official state documents, the open access platform for those with publicly available CVs and news articles from newspapers, magazines and blogs. Party affiliations and data on the secretary and the governor’s party and the alignment of these figures with the government of Jair Bolsonaro were confirmed by consulting data published by the Center for Politics and Economics of the Public Sector of the Getulio Vargas Foundation. For each secretary, we recorded whether



Source: Authors’ own work

Figure 1.
Theoretical
framework: the
contribution of
relational and
reputational shocks
to leadership
turnover

they held a management or health degree and if he or she had administrative experience in public health, management or other types of business training.

Based on the cited sources, we coded reasons for the removal of the secretary as either relational or reputational shocks. Secretaries who were removed or resigned because of a relational shock include those for which we could identify:

- political disagreements;
- resignation for executive or legislative office;
- return of the previous secretary following an electoral defeat; or
- interim tenure as the reason for turnover.

A reputational shock was defined as cases with corruption or fraud allegations. We also indicated the secretaries who left the office for personal, other or non-informed reasons that led to the public official's replacement. Those for whom no reason could be identified were classified as personal or other reasons. The data analysis compares the overall duration of state health secretaries in office with those who left for political disagreements (relational reason) and accusations of corruption, fraud and ethical violations (reputational reason).

To capture the health emergency in each state and its evolution over the study period, we collected data on infected cases and deaths by COVID-19 based on official information released by the state health officials available in a public data repository ([Brasil.io, 2023](#)). The first confirmed SARS-CoV-2 case in Brazil was on February 26, 2020. We collected daily up until the end of the study period (December 31, 2022). It should be noted, however, that the reported number of cases varied with the intensity of testing policies across states during the health emergency ([Barberia et al., 2023](#)).

The data consists of a time-series pooled design ($n = 27$ states and the federal district and $t = 1,059$ days for the COVID-19 pandemic). The dependent variable is the turnover status of each health secretary in the health emergency. Since the first case of COVID-19 in Brazil was registered on February 24, 2020, we analyzed the turnover patterns of health secretaries during the pandemic, including those who were in office as of February 1, 2020, and those appointed after this date up until December 31, 2022. Given the study's objective, we did not analyze health secretaries who left office from the start of the gubernatorial term, January 2019, to before the onset of the crisis, January 2020 (before the health emergency). The data set includes the specific day, month and year the 75 health secretaries took and left office. The database is organized as multiple record data. There is a line of data for each health secretary who held office to capture the covariates that, although constant within the specific term, change from one secretary to another.

The research analysis was undertaken in two stages. First, a descriptive analysis of health secretary turnover and correlations with the factors contributing to shorter tenure durations was undertaken. In the second stage, the hypotheses were tested using Cox duration models ([Ruhe, 2018](#)). The Cox model assumes that covariate effects are proportional to the hazard rate over time. Considering the variables in which the assumption that proportional hazards did not hold, we estimated Cox survival functions with time-varying coefficients ([Ruhe, 2016](#)).

Results

There was a marked rise in the volatility of the duration of health secretaries at the state level in the Brazilian federation during the COVID-19 pandemic. Between February 2020 and December 2022, 75 health secretaries oversaw efforts to combat the pandemic, and their tenure lasted 517 days, on average. In this section, we provide evidence that confirms

hypotheses *H1* and *H2* of our theoretical framework. In addition to detailing the contribution of relational and reputational shocks in influencing the leadership turnover of the state health secretary, we show that there are important differences across states of the federation.

The duration of the health secretary in his or her post during the COVID-19 pandemic varied greatly. In some states, the health secretary lasted less than one year: Roraima (151 days), Santa Catarina (216), Distrito Federal (242), Amazonas (270) and Rio de Janeiro (291) (Figure 2, Panel A). In contrast, in the states of Espírito Santo, Pernambuco, Rio Grande do Norte and Rio Grande do Sul, the secretary of health served the entire governor's tenure term (1,460 days) (see Figure 2, Panel B). As we explain in further detail in the following paragraphs, there is robust support that the duration of a health secretary's term is influenced by *H1* more than *H2* during COVID-19.

In the initial year of the COVID-19 health emergency, 17 turnovers occurred across nine states in 2020, as depicted in Figure 3. The primary cause for these removals was relational, accounting for nine cases, while an alarming six health secretaries were replaced due to reputational concerns. The highest number of corruption allegations and fraud accusations occurred in the first year of the pandemic. Notably, this year witnessed substantial increases in unplanned healthcare expenditures, primarily toward building emergency hospitals and acquiring equipment such as ventilators and ICU hospital beds. Many reputation-driven removal cases involved health secretaries that experienced a crisis related to these contracts and accusations of overpricing or fraud. It is worth highlighting that in four states (Roraima, the Federal District, Amazonas and Rio de Janeiro), the position of health secretary changed hands three times in twelve months, underscoring the instability and challenges faced in ensuring consistent leadership in these states during the first year of the health emergency.

Turnover trends continued to persist in the subsequent two years of the pandemic, mirroring the pattern observed in 2020. Once again, policy differences remained a key driver of turnover rates. In 2021, 10 states witnessed changes in their health secretaries, with two states (Roraima and Santa Catarina) experiencing two substitutions. Most of these turnovers ($n = 9$) occurred due to resignations or replacements motivated by relational factors. However, one secretary (in the state of Amazonas) departed for reputational reasons, while two others (in Acre and Ceará) cited personal reasons for their exit. In 2022, many replacements were driven by relational factors ($n = 16$), but there were also departures for

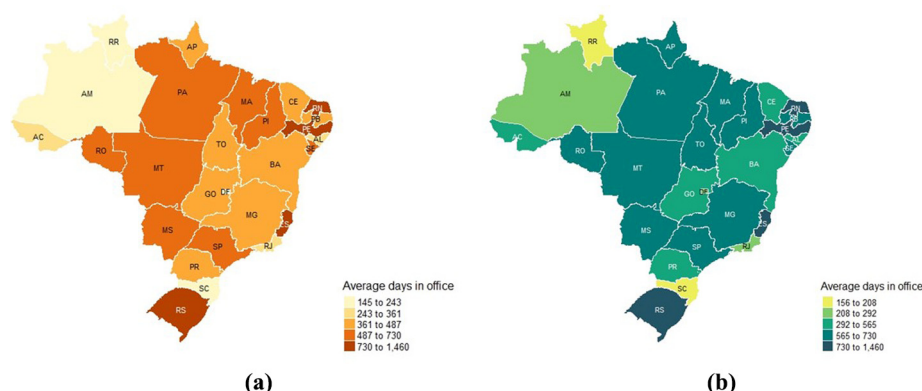


Figure 2.
(a) Average days in
office for a state
health secretary from
January 2019 to
December 2022. (b)
Average days in
office for a state
health secretary from
February 2020 to
December 2022

Source: Authors' own work



Figure 3. COVID-19 pandemic-related replacements of health secretaries in 26 states and the federal district, categorized by reason for removal

Source: Authors’ own work

personal reasons ($n = 3$). Notably, four states (Alagoas, Goiás, Paraná and Santa Catarina) experienced more than one change in the position of health secretary in 2022.

It is worth noting that elections were an important driver of turnover patterns. Municipal elections were held in November 2020, and state and federal elections in October 2022. There were 17 replacements in 2020 and 19 in 2022. The volatility in these periods is partially driven by the departures of secretaries who left to run for office. Some of these same individuals returned to their posts after suffering electoral defeat.

Figure 2 (Panel B) also reveals that the states witnessing the highest turnovers in health leadership during COVID-19 were concentrated in the North, Southeast and South regions of the country. To put these turnover rates in context, from 1995 to 2018, the average number of changes in health secretary appointments stood at 2.3, indicating that a governor would typically replace health secretaries twice within a four-year mandate. However, in the first three years of the COVID-19 pandemic, an average of 2.7 health secretaries served at the state level. This finding is particularly significant as it pertains to a period shorter than the entire gubernatorial term, signifying a heightened level of turnover volatility throughout the pandemic.

Furthermore, this heightened pattern of turnover occurred across the federation. Among 27 federative units (26 states and one federal district), 23 changed health secretaries at least once during the pandemic. In 10 of these states, governors made more than one change; five states made two changes, and two states made four changes. The Federal District made five replacements, and Roraima and Santa Catarina made six changes. It is worth noting that

overall, Roraima had the highest number of substitutions between January 2019 and December 2022, with ten changes. Of these, seven were during the health emergency. Figure 3 shows that 59 replacements occurred between January 2019 and December 2022. Of these, 48 (81.4%) were during COVID-19.

Table 1 confirms that relational rather than reputational factors drove health secretary turnover throughout the COVID-19 pandemic. Public health executives were removed from their posts primarily due to political disagreements and dynamics (interim tenure and return of the previous secretary). Accusations of ethical violations and/or fraud were the cause of turnover in fewer cases. In total, 34 secretaries were removed for relational, and eight for reputational reasons. Moreover, secretaries who left because they had disagreements regarding health policies were nevertheless more likely to remain in office (505 days in office, on average) than those who left for reputational reasons (359d days in office, on average).

Table 2 summarizes the mean duration and total number of health secretaries who served throughout the COVID-19 pandemic. Notably, while it is not obligatory by regulation, most governors preferred appointing health secretaries with academic training in health-related disciplines, including medicine, nursing, biomedicine, dentistry and psychology. This trend was further reflected in the fact that 67% of health secretaries during the pandemic possessed an academic background in health. Furthermore, a significant proportion (85.3%) of these secretaries had prior experience within the Brazilian public health system, either as healthcare managers or as professionals engaged in patient care. Many of them acquired this experience through their academic pursuits in health-related fields, which allowed them to develop practical expertise within the public healthcare system.

Health secretaries with professional expertise or an academic degree in public health exhibited a longer average tenure (560 days) during the pandemic than those with no experience (375 days). Considering academic training in management, 46.5% of the health secretaries in the office had at least one degree in public administration, management or health systems management, and 49.3% during the pandemic. Surprisingly, health secretaries lacking management experience had a longer average tenure (658 days).

Table 3 presents the Cox models estimates of the effect of covariates on the duration until a secretary exits the state health secretariat. The model in column 1 of Table 3 uses several predictor variables to capture the gender of the secretary, her educational level, management education, management work experience, public health education and public health experience

Reason for removal	No. of secretaries	Mean (days)	Standard deviation	Minimum	Maximum
Relational	34	505	460	12	1,185
Reputational	8	359	253	79	757
Mean number of days(minimum and maximum)			<i>Relational</i>		<i>Reputational</i>
Management experience					
Professional background in public management			495 (12–1,185)		339 (79–757)
Academic degree in management			348 (12–1,184)		271 (79–757)
Public health experience			<i>Relational</i>		<i>Reputational</i>
Professional background in public health			550 (12–1,185)		339 (79–757)
Academic degree in a health-related field			560 (12–1,185)		479 (88–757)

Source: Authors' own work

Table 1.
Duration in office for
health secretaries
exiting due to
relational and
reputational reasons
in 26 states and the
federal district
during COVID-19

and two dichotomous variables that capture whether the secretary who left office departed for reputational or relational reasons. Following Schoenfeld residuals tests on all predictor variables, relational conflict was interacted with a linear function of time and these are also reported.

The model results reported in column 2 of Table 3 uses the same predictor variables and an additional predictor variable for the COVID-19 deaths per 100,000 observed in the state in the epidemiological week in which the health secretary’s term ended. Following Schoenfeld residuals tests on all predictor variables, relational conflict and COVID-19 deaths were interacted with a linear function of time.

Panels A and B of Figure 4 depict the probability of survival for those who did not suffer any relational or reputational shock and those who departed due to a relational or reputational shock. Both figures suggest that there is strong support for H1 and H2. In the

Table 2.
Public health and management experience of health secretaries in 26 states and the federal district during COVID-19 and days in office

Secretary traits	No. of secretaries	Mean (days in office)	SD (days in office)	Minimum (days in office)	Maximum (days in office)
Men	63	536	439	12	1,460
Women	12	413	408	37	1,460
<i>Public health experience</i>					
Academic degree in health-related field	50	560	423	12	1,460
Professional background in public health	64	541	440	12	1,460
No experience	7	375	438	31	1,185
<i>Management experience</i>					
Academic degree in management	37	438	395	12	1,460
Professional background in public management	53	503	426	12	1,460
No experience	14	658	482	34	1,460
Source: Authors’ own work					

Table 3.
Cox models with time-varying adjustments

Explanatory variables	Time-varying adjustment for relational conflict	Time-varying adjustment for relational conflict and COVID-19 deaths
Reputational	2.168*** (3.98)	1.203* (2.34)
Relational	2.484*** (3.57)	2.399** (2.74)
Academic health background	−0.231 (−0.46)	−0.573 (−1.02)
Academic management background	0.697 (1.68)	0.983* (2.53)
Management work expertise	0.0443 (0.1)	0.325 (0.71)
Public health work expertise	−0.591 (−1.03)	−0.448 (−0.85)
Education (categorical variable)	−0.198 (−1.21)	−0.231 (−1.45)
Male	−0.713 (−0.90)	−1.919* (−2.52)
COVID-19 Deaths		−0.0708* (−2.06)
<i>Time-varying coefficients</i>		
Relational	−0.000822 (−0.83)	−0.000191*** (−3.95)
COVID-19 deaths		−0.00492*** (−4.39)
N	75	75
Notes: <i>t</i> statistics in parentheses; * <i>p</i> < 0.05, ** <i>p</i> < 0.01, *** <i>p</i> < 0.001		
Source: Authors’ own work		

pre-pandemic period, Panel A confirms that the duration of a secretary was reduced for public health secretaries who either experienced a reputational or relational conflict in their state. All else constant and compared to the group of secretaries who remained in office for most of their four-year terms, the predicted probability of survival for health secretaries is considerably lower.

Panel B reports the probability of remaining in office considering time-varying adjustments in the Cox model for COVID-19 deaths and relational conflicts. As this figure shows, there are marked differences in the likelihood of survival once we consider COVID-19 dynamics. In the pre-pandemic period, health secretaries who experienced a relational conflict were far less likely to remain in office. Still, as time progressed, the likelihood of their survival remained unchanging and hovering slightly below 40%. In contrast, secretaries who experienced a reputational conflict were much more likely to be removed once the COVID-19 pandemic worsened in the state. In other words, worsening COVID-19 dynamics were more likely to increase the dismissal of health secretaries for mismanagement, including corruption allegations.

Caution should be exercised when interpreting these findings due to several limitations. Firstly, we identified the main reason for the secretary's departure. However, it is often the case that secretaries leave for reputational and relational reasons or that a secretary will cite a personal reason for exiting when there is a more substantive factor driving the leader's departure. Secondly, the prevalence of health secretaries with combined public health and management experience makes it challenging to ascertain the individual contributions of each factor to the duration of a secretary's tenure.

Discussion

As the COVID-19 pandemic represents a moment of crisis, not only in health, but also in economic, social and political terms, the management of this emergency was impacted by the stability in the decision-making of public managers in the health sector (Barberia *et al.*, 2021b). In moments of crises, such as a pandemic, the turnover of leadership positions has implications for crisis management (Boin and Hart, 2003; Halverson *et al.*, 2021). The transition between health leaders can delay timely responsiveness during a critical moment. Thus, we urgently need to understand the internal and external factors that pressure health officials to leave office during crises.

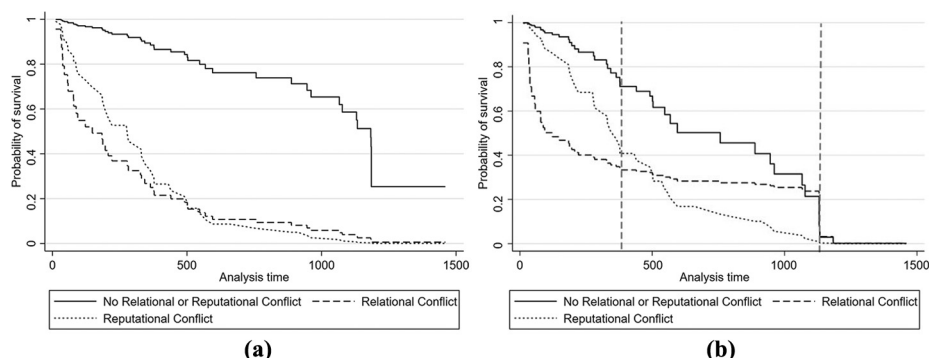


Figure 4.
(a) Cox model with time-varying adjustment for relational conflict (b).
(b) Cox model with time-varying adjustment for relational conflict and COVID-19 deaths per 100,000

Source: Authors' own work

Our results indicate that possible conflicts between the manager and his direct supervisor, such as the state governor, result in relational difficulties of permanence in the position. They are the main reason for the health secretary's removal in Brazilian states during the COVID-19 pandemic. Likewise, but to a lesser extent, accusations of fraud result in the dismissal of public managers from commissioned positions. Conflicts with elected officials were also a cause for turnover among state and territorial health officials in the USA during the first year of the COVID-19 pandemic (Halverson *et al.*, 2021).

The unified public health system and past successes with epidemic control placed South America's largest federation in an advantageous position to respond effectively to the SARS-CoV-2 emergency. However, there were significant disagreements between elected officials and senior public health managers on health policies since the onset of the COVID-19 pandemic. Furthermore, as the COVID-19 pandemic persisted for several years, these conflicts resulted in turnover at the ministry and state levels.

At the federal level, Jair Bolsonaro had the highest turnover in the health ministry of any democratically elected president since 1988. In April 2020, Bolsonaro fired his Minister of Health, Luiz Henrique Mandetta, after disagreeing over physical distancing policies. His next Minister of Health, Nelson Teich, resigned from office by mid-May following policy disagreements and refusing to yield to Bolsonaro's pressures concerning the use of hydroxychloroquine as a COVID-19 treatment. Bolsonaro then appointed Eduardo Pazuello, an Army General with no medical experience, as the interim Minister of Health from June 2 to September 16, 2020, a post he subsequently held until March 23, 2021, when a cardiologist, Mauricio Queiroga, was appointed as Bolsonaro's last health minister. Thus, our study shows a higher turnover during the COVID-19 period compared with the SARS emergency in 2007. Thus, the volatility of health secretary turnover at the state level in 2020–2022 coincided with the turnover at the federal level.

Our study suggests that secretaries with prior experience in public health, mainly with academic training related to health, remained in office longer during the pandemic compared to those with no familiarity. Having an academic degree in public health was an essential mediator in extending the days in office of the state secretaries in charge of the health of their respective populations. The results indicate that 49.3% of health secretaries who worked during the pandemic have academic management training, which is considerably higher than the findings for the USA. Halverson *et al.* (2021) found that most public health officials in the USA who held state office were physicians (64.6%), but only 22% had an academic background in management or administration. The findings suggest that skills in operational management, strategic thinking and change management, which are the types of training not covered in public health or medical schools, are also being incorporated into decision-making at the state level in Brazil.

Our findings have important implications for future research. Qualitative interviews with a representative sample of health secretaries who presided over health emergencies at the federal, state and municipal levels in Brazil, including those who left office during the crisis due to a reputational or relational shock could help to elucidate the relative importance of these drivers on turnover. Furthermore, this study has analyzed turnover patterns during COVID-19. Understanding if these patterns were also observed in other health emergencies should be on the research agenda of those scholars seeking to understand how leadership can enhance or limit an effective health emergency response.

Practical implications

This study contributes to the growing literature on pandemic preparedness and the critical role of leadership and collaborative crisis management (Geerts *et al.*, 2021; Margherita *et al.*,

2021). The research findings presented in this study have important implications for organizations and health systems in countries such as Brazil, with large, decentralized public health systems. Evidently, no government was adequately prepared for such a catastrophic event; however, the high volatility in leadership undoubtedly contributed to suboptimal governance in the public health sector in Brazil.

The patterns identified in this study help explain why Brazil has experienced such severe repercussions from the pandemic. Specifically, our findings underscore the influential role of reputational and relational factors in driving the turnover of state health secretaries. We show that these pressures increase during the acute phase of a crisis. These insights contribute to a better understanding of the challenges faced in effectively managing public health crises and emphasize the need for strategic interventions to enhance leadership stability and governance in the face of future health crises.

Since the study tracks the early and more prolonged phases of managing the COVID-19 health emergency, our research also provides compelling evidence of the persistent challenges posed by high turnover rates in health leadership. As we show, not only is the volatility in leadership a significant obstacle that can disrupt efforts and impede the timely alignment of health policies and guidelines between state and central governments, but its frequency can increase as the crisis worsens. Furthermore, electoral cycles, which contribute to increased turnover among health leaders seeking elected offices, can exacerbate turnover.

The findings also suggest that efforts to combat corruption and fraud in the health sector may contribute to reducing turnover in health secretary leadership. Thus, the research highlights the urgent need for effective plans to address the challenge of leadership turnover in crisis management. Such strategies should consider relational and reputational shocks, the impact of electoral cycles and the promotion of transparency and accountability in the health sector. Reputational shocks seem to be worse than relational ones because, in these cases, leaders have more difficulties operating the “revolving door” in the public sphere as there is greater scrutiny. Leadership in this sphere must consider facilitating constructive dialogues between health secretaries and governors to improve emergency responses in the face of health, economic and political crises like the COVID-19 pandemic.

References

- Alexander, J.A., Lee, S.Y.D. and Bazzoli, G.J. (2003), “Governance forms in health systems and health networks”, *Health Care Management Review*, Vol. 28 No. 3, pp. 228-242.
- Allen, P., Jacob, R.R., Lakshman, M., Best, L.A., Bass, K. and Brownson, R.C. (2018), “Lessons learned in promoting evidence-based public health: perspectives from managers in state public health departments”, *Journal of Community Health*, Vol. 43 No. 5, pp. 856-863.
- Anderson, R.A. and McDaniel, R.R. Jr (2000), “Managing health care organizations: where professionalism meets complexity science”, *Health Care Management Review*, Vol. 25 No. 1, pp. 83-92.
- Andrus, J.L., Withers, M.C., Courtright, S.H. and Boivie, S. (2019), “Go your own way: exploring the causes of top executive turnover”, *Strategic Management Journal*, Vol. 40 No. 7, pp. 1151-1168.
- Barberia, L., Costa, S.F. and Sabino, E.C. (2021a), “Brazil needs a coordinated and cooperative approach to tackle COVID-19”, *Nature Medicine*, Vol. 27 No. 7, pp. 1133-1134.
- Barberia, L., Plumper, T. and Whitten, G.D. (2021b), “The political science of Covid-19: an introduction”, *Social Science Quarterly*, Vol. 102 No. 5, pp. 2045-2054.
- Barberia, L., Boing, A., Gusmão, J., Miyajima, F., Abud, A., Kemp, B., Zamudio, M. and Moraes de Sousa, T.C. (2023), “An assessment of the public health surveillance strategy based on molecular

- testing during three major pandemic waves of COVID-19 in Brazil”, *PLOS Global Public Health*, Vol. 3 No. 8, p. e0002164, doi: [10.1371/journal.pgph.0002164](https://doi.org/10.1371/journal.pgph.0002164).
- Beck da Silva Etges, A.P., Cardoso, R.B., Marcolino, M.S., Ruschel, K.B., Coutinho, A.P., Pereira, E.C., Anschau, F., Aranha, F., Carrilho, F., Vietta, G. and Bastos, G.A.N. (2021), “The economic impact of COVID-19 treatment at a hospital-level: investment and financial registers of Brazilian hospitals”, *Journal of Health Economics and Outcomes Research*, Vol. 8 No. 1, pp. 36-41.
- Bevan, G., Evans, A. and Nuti, S. (2019), “Reputations count: why benchmarking performance is improving health care across the world”, *Health Economics, Policy, and Law*, Vol. 14 No. 2, pp. 141-161.
- Boin, A. and 't Hart, P. (2010), “Organising for effective emergency management: lessons from research”, *Australian Journal of Public Administration*, Vol. 69 No. 4, pp. 357-371.
- Boin, A. and Hart, P.T. (2003), “Public leadership in times of crisis: mission impossible?”, *Public Administration Review*, Vol. 63 No. 5, pp. 544-553.
- Boin, A., 't Hart, P., Stern, E. and Sundelius, B. (2005), *The Politics of Crisis Management: Public Leadership under Pressure*, Cambridge University Press, Cambridge.
- Bousquat, A., Giovanella, L., Facchini, L.A., de Mendonça, M.H.M., Nedel, F.B., Cury, G., dos Santos Mota, P.H., Schenkman, S., Chueiri, P.S. and Alves, M.C.G.P. (2022), “The Brazilian PHC response to the COVID-19 pandemic: individual and collective comprehensiveness and performance profiles”, Research Square, doi: [10.21203/rs.3.rs-1909134/v1](https://doi.org/10.21203/rs.3.rs-1909134/v1).
- Brasil.io (2023), “Dados diários mais recentes do coronavírus por município brasileiro”, available at: <https://brasil.io/covid19/> (accessed 1 March 2023).
- Camerlo, M. and Perez-Liñan, A. (2015), “The politics of minister retention in presidential systems: technocrats, partisans, and government approval”, *Comparative Politics*, Vol. 47 No. 3, pp. 315-333.
- Cortez, C., Mansour, O., Qato, D.M., Stafford, R.S. and Alexander, G.C. (2021), “Changes in short-term, long-term, and preventive care delivery in US office-based and telemedicine visits during the COVID-19 pandemic”, *JAMA Health Forum. jamanetwork.com*, Vol. 2 No. 7, p. e211529.
- Cregård, A., Corin, L. and Skagert, K. (2017), “Voluntary turnover among public sector managers: a review”, *Scandinavian Journal of Public Administration*, Vol. 21 No. 2, pp. 89-114.
- Cruz, I. (2020), “Público e privado: a disputa por leitos de UTI na pandemia”, Nexo, available at: www.nexojornal.com.br/expresso/2020/05/06/P%C3%BAblico-e-privado-a-disputa-por-leitos-de-UTI-na-pandemia (accessed 3 March 2023).
- Ezzati, F., Mosadeghrad, A.M. and Jaafari-pooyan, E. (2023), “Resiliency of the Iranian healthcare facilities against the covid-19 pandemic: challenges and solutions”, *BMC Health Services Research. Springer*, Vol. 23 No. 1, p. 207.
- Fonseca, E.M.D., Shadlen, K.C. and Bastos, F.I. (2021), “The politics of COVID-19 vaccination in middle-income countries: lessons from Brazil”, *Social Science and Medicine*, Vol. 281, p. 114093.
- Fraser, M., Castrucci, B. and Harper, E. (2017), “Public health leadership and management in the era of public health 3.0”, *Journal of Public Health Management and Practice*, Vol. 23 No. 1, pp. 90-92.
- Geerts, J.M., Kinnair, D., Taheri, P., Abraham, A., Ahn, J., Atun, R., Barberia, L., Best, N.J., Dandona, R., Dhahri, A.A. and Emilsson, L. (2021), “Guidance for health care leaders during the recovery stage of the COVID-19 pandemic: a consensus statement”, *JAMA Network Open*, Vol. 4 No. 7, p. e2120295.
- Ginter, P.M., Jack Duncan, W. and Swayne, L.E. (2018), *The Strategic Management of Health Care Organizations*, John Wiley and Sons, Hoboken, New Jersey.
- Gomes, S., Lacerda, A.D.F.D. and Silva, A.L.N.D. (2021), “Critérios técnicos, políticos e dinâmica regional na nomeação de secretários estaduais: um estudo de caso do Rio Grande do Norte”, *Revista de Sociologia e Política*, Vol. 28 No. 76, p. e005.
- Haldane, V., De Foo, C., Abdalla, S.M., Jung, A.S., Tan, M., Wu, S., Chua, A., Verma, M., Shrestha, P., Singh, S. and Perez, T. (2021), “Health systems resilience in managing the COVID-19 pandemic: lessons from 28 countries”, *Nature Medicine*, Vol. 27 No. 6, pp. 964-980.

-
- Halverson, P.K., Yeager, V.A., Menachemi, N., Fraser, M.R. and Freeman, L.T. (2021), "Public health officials and COVID-19: leadership, politics, and the pandemic", *Journal of Public Health Management and Practice*, Vol. 27 No. Supplement 1, pp. S11-S13.
- Haque, A. (2021), "The COVID-19 pandemic and the role of responsible leadership in health care: thinking beyond employee well-being and organisational sustainability", *Leadership in Health Services*, Vol. 34 No. 1, pp. 52-68.
- Kooli, C. (2021), "COVID-19: public health issues and ethical dilemmas", *Ethics, Medicine, and Public Health*, Vol. 17, p. 100635.
- Margherita, A., Elia, G. and Klein, M. (2021), "Managing the COVID-19 emergency: a coordination framework to enhance response practices and actions", *Technological Forecasting and Social Change*, Vol. 166, p. 120656.
- Massuda, A., Hone, T., Leles, F.A.G., De Castro, M.C. and Atun, R. (2018), "The Brazilian health system at crossroads: progress, crisis and resilience", *BMJ Global Health*, Vol. 3 No. 4, p. e000829.
- Park, S. and Fowler, L. (2021), "Political and administrative decentralization and responses to COVID-19: comparison of the United States and South Korea", *International Journal of Organization Theory and Behavior*, Vol. 24 No. 4, pp. 289-299.
- Pereira, A.K., Oliveira, M.S. and Sampaio, T.D.S. (2020), "Asymmetries of state government social distancing policies in the face of COVID-19: political and technical-administrative aspects", *Revista ADM*, Vol. 54 No. 4, pp. 678-696.
- Ramírez de la Cruz, E.E., Grin, E.J., Sanabria-Pulido, P., Cravacuore, D. and Orellana, A. (2020), "The transaction costs of government responses to the COVID-19 emergency in Latin America", *Public Administration Review*, Vol. 80 No. 4, pp. 683-695.
- Ruhe, C. (2016), "Estimating survival functions after Stcox with time-varying coefficients", *The Stata Journal: Promoting Communications on Statistics and Stata*, Vol. 16 No. 4, pp. 867-879.
- Ruhe, C. (2018), "Quantifying change over time: interpreting time-varying effects in duration analyses", *Political Analysis*, Vol. 26 No. 1, pp. 90-111.
- Siddiqi, S., Masud, T.I., Nishtar, S., Peters, D.H., Sabri, B., Bile, K.M. and Jama, M.A. (2009), "Framework for assessing governance of the health system in developing countries: gateway to good governance", *Health Policy*, Vol. 90 No. 1, pp. 13-25.
- Zahra, S.A., Priem, R.L. and Rasheed, A.A. (2007), "Understanding the causes and effects of top management fraud", *Organizational Dynamics*, Vol. 36 No. 2, pp. 122-139.

About the authors

Lorena G. Barberia is an Associate Professor in the Department of Political Science at the University of Sao Paulo. Lorena G. Barberia is the corresponding author and can be contacted at: lorenabarberia@usp.br

Gilmar Masiero is an Associate Professor in the Management Department at the University of Sao Paulo.

Iana Alves de Lima is a PhD candidate in Public Administration and Government Ph.D. Program at the School of Business Administration at the Fundação Getulio Vargas.

Luciana Santana is an Associate Professor at the Institute of Social Science at the Federal University of Alagoas.

Tatiane C. Moraes de Sousa is a Post-Doctoral Fellow and Instructor in the Department of Political Science at the University of Sao Paulo.

For instructions on how to order reprints of this article, please visit our website:

www.emeraldgrouppublishing.com/licensing/reprints.htm

Or contact us for further details: permissions@emeraldinsight.com